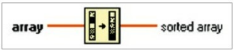
block diagram using graphical representations of functions to control front panel objects. Code on the block diagram is graphical code, also known as G code or block diagram code.

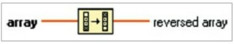
Implementation of Shannon's Fano Elias algorithm in LabVIEW requires the following components:

Sort 1D array. This function returns a sorted version of the array with the elements arranged in ascend-



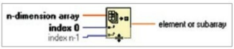
ing order. If the array is an array of clusters, the function sorts the elements by comparing the first elements. If the first elements match, the function compares the second and subsequent elements. The connector pane displays default data types for this polymorphic function.

Reverse 1D array. This function reverses the order of elements in the array, where the array can be of any



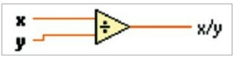
type. The connector pane displays default data types for this polymorphic function.

Index array. This function returns the element or sub-array of n-dimension array at index. When you wire an array to this function, the function resizes automatically to display index inputs for each dimension in the array you wire to n-dimension array.



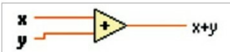
You can also add additional elements or sub-array terminals by resizing the function. The connector pane displays default data types for this polymorphic function.

Divide. If you wire two waveform values or two dynamic data type values to this function, error in and



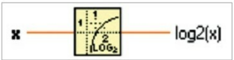
error out terminals appear on the function. The connector pane displays default data types for this polymorphic function.

Add. If you wire two waveform values or two dynamic data type values to this function, error in and error out terminals appear on the func-



tion. You cannot add two time-stamp values together. Dimensions of two matrices that you want to add must be the same. Otherwise, this function returns an empty matrix. The connector pane displays default data types for this polymorphic function.

Logarithm base 2. If x is 0, log2(x) is negative infinity. If x is not complex and is less than 0, log2(x) is NaN. The

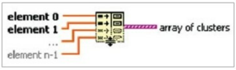


connector pane displays default data types for this polymorphic function.

Round toward +infinity. For example, if input is 3.1, the result is 4. If input is -3.1, the result is -3. The

connector pane displays default data types for this polymorphic function.

Build cluster array. This function bundles each element input into a cluster and assembles all element



clusters into an array of clusters. The connector pane displays default data types for this polymorphic function.

Block diagram description

Step 1. Begin with the decimal fraction and multiply by 2. The whole number part of the result is the first binary digit to the right of the point.

Step 2. Disregard the whole number part of the previous result and multiply the rest again. The whole number part of this new result is the second binary digit to the right of the point. Continue the process until you get 0 as the decimal part, or until you recognise an infinite repeating pattern.

Step 3. Disregard the whole number part of the previous result and multiply by 2 once again. The whole number part of the result is now the next binary digit to the right of the point. Continue the process according to the length.

Testing procedure

1. Download and install LabVIEW.
2. Click LabVIEW on desktop.
3. Open Shannonfanoeliasfinal.vi
4. Provide source symbols for x1, x2, etc, probabilities P1, P2, etc. While applying the input probability value, the sum will be equal to 1, since the total probability of symbol should be 1.
5. Click Run. Generated code should satisfy prefix and instantaneous properties.

Note. Shannonfanoelias.vi is the main VI. We have binary converter.vi along with this main VI. While opening the main VI, if you face any error or question mark, double-click on the question mark. The error will be moved since it is linked with binary converter.vi. EFY

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

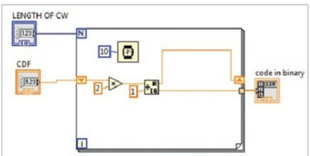


Fig. 3: Decimal-to-binary conversion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